

Decision Tree Based Machine Learning Framework for Early Risk Prediction in Soldier Health Monitoring Using IoT and LoRa Communication

M A Seshasaai
*Department of Electronics and
Communication Engineering
R.M.K. Engineering College
Tiruvallur, India
animoanimations3@gmail.com*

Sai Janavi P H
*Department of Electronics and
Communication Engineering
R.M.K. Engineering College
Tiruvallur, India
saijanaviph@gmail.com*

K Bala Manikanta
*Department of Electronics and
Communication Engineering
R.M.K. Engineering College
Tiruvallur, India
kallebalamanikanta30@gmail.com*

Kamaleshwaran M*
*Department of Electronics and
Communication Engineering
R.M.K. Engineering College
Tiruvallur, India
klr969496@gmail.com*

Lakshana R
*Department of Electronics and
Communication Engineering
R.M.K. Engineering College
Tiruvallur, India
rk.lakshana2005@gmail.com*

S Vijayalakshmi
*Department of Electronics and
Communication Engineering
R.M.K. Engineering College
Tiruvallur, India
svl.eie@rmkec.ac.in*

Abstract—Soldiers deployed in extreme climatic border regions such as Jaisalmer (Rajasthan) and Leh (Ladakh) are exposed to severe heat, extreme cold, high physical stress, and limited medical support. Continuous monitoring of physiological and environmental parameters is essential for ensuring soldier safety and timely medical intervention. This paper presents a Decision Tree-based soldier health monitoring and early risk prediction system for real-time assessment using machine learning. The proposed system utilizes multi-sensor data including heart rate (BPM), SpO_2 , temperature, humidity, and three-axis motion (AX, AY, AZ) to classify soldier health into eight states: Normal, Fatigue, Heat_Risk, Cold_Risk, Panic_Risk, Injury_Risk, Heat_Stroke, and Severe_Cold_Risk. The hardware architecture consists of a transmitter unit with MAX30102, MPU6050, DHT22, GPS NEO-6M, LoRa Ra-02, and ESP32, and a receiver unit with LoRa Ra-02, ESP32, and a laptop-based dashboard. Data are transmitted via LoRa communication without GSM or internet dependency. A Decision Tree model trained on collected and logically generated datasets achieved 99.56% accuracy. The Flask-based web dashboard provides real-time visualization, health prediction, GPS tracking, and historical storage using SQLite. Experimental results demonstrate the feasibility of integrating machine learning with long-range wireless communication for predictive soldier health monitoring in harsh environments.

Keywords—Decision Tree, soldier health monitoring, early risk prediction, LoRa communication, ESP32, machine learning, health risk assessment, GPS tracking.

I. INTRODUCTION

Military personnel deployed in border regions such as Jaisalmer (Rajasthan) and Leh (Ladakh) operate under extreme environmental and physiological stress conditions, including high temperature, severe cold, low oxygen levels, physical exertion, and psychological strain. These factors can lead to critical health issues such as heat stroke, hypothermia, fatigue, panic, and injury. Existing soldier monitoring systems primarily focus on real-time physiological monitoring and threshold-based alert generation. However, as highlighted in prior studies, these approaches are largely reactive and lack predictive intelligence, with most works relying on fixed thresholds or rule-based decision-making, while AI/ML-based health-risk prediction remains an open research gap [1], [2].

To address this limitation, this paper proposes a Decision Tree-based soldier health monitoring and early risk prediction system. The system integrates heart rate and SpO_2 (MAX30102), temperature and humidity (DHT22), motion sensing (MPU6050), and GPS location (NEO-6M) to analyze soldier health conditions. The trained Decision Tree model classifies eight health states—Normal, Fatigue, Heat_Risk, Cold_Risk, Panic_Risk, Injury_Risk, Heat_Stroke, and Severe_Cold_Risk—with an achieved accuracy of 99.56%. The system architecture consists of an ESP32 and LoRa-based transmitter and receiver setup for long-range communication [2], [8], [9]. Data transmitted via LoRa are processed through a Flask-based web dashboard that enables real-time monitoring, GPS visualization, health prediction, graphical analysis, and historical storage using SQLite.

II. LITERATURE REVIEW

Recent research on soldier health monitoring systems has primarily focused on real-time monitoring, location tracking, and emergency alert generation. However, the reviewed literature reveals that these systems generally rely on threshold-based monitoring or simple alert mechanisms and do not incorporate artificial intelligence or machine learning for predictive health-risk assessment.

Thotakura *et al.* [1] proposed the LifeLine system, which utilized ESP8266, IoT integration, and GSM-based communication for soldier health monitoring and situational awareness. The system monitored physiological parameters and generated alerts; however, it did not employ AI or machine learning for health-risk prediction. The authors explicitly identified AI and machine learning-based predictive health assessment as future work.

Umamaheswari *et al.* [2] presented a military monitoring system integrating LoRa communication and image-processing technologies to enhance situational awareness. Their work demonstrated the effectiveness of LoRa for long-range communication in military environments. Nevertheless, the system focused on monitoring and surveillance rather than predictive health-risk analysis. Similar to [1], the authors suggested future integration of AI and machine learning technologies.

Elakya *et al.* [3] proposed an IoT-based smart human safety device using ESP32, physiological sensors, GPS, GSM, and a Random Forest algorithm for anomaly detection and automated emergency response. Although the system incorporated machine learning for general human health monitoring, it relied on GSM and cloud connectivity and was not designed for soldier-specific health-risk prediction in remote military environments using LoRa communication.

Aggarwal *et al.* [4] developed a soldier health and position tracking system capable of transmitting pulse rate, oxygen level, environmental information, and location data to a control center. The system improved communication, monitoring, and emergency response capabilities but did not provide predictive health-state classification.

Patel *et al.* [5] implemented a real-time soldier health and position monitoring system incorporating heart-rate sensing, body-temperature monitoring, GPS tracking, emergency push-button alerts, and an LCD interface. Experimental results demonstrated reliable monitoring and emergency-response functionality. However, the system remained limited to data monitoring and alert generation without predictive intelligence.

Pund *et al.* [6] proposed an IoT-based E-vest system for soldier health monitoring and position tracking using wearable sensors. The system continuously collects and transmits physiological and location data, enabling real-time monitoring and alert generation for abnormal conditions. However, it focuses on monitoring and alerts rather than machine learning-based predictive health-risk assessment, which remains a scope for further improvement.

Swetha Priyanka *et al.* [7] proposed a GPS- and GSM-based soldier safety system that monitored pulse rate, body temperature, and location information. Sensor data were stored in the ThingsBoard cloud platform for analysis, and manual emergency buttons were provided for soldier assistance. Although cloud-based monitoring was achieved, the system lacked machine-learning-based health-risk prediction.

Sharmila Devi *et al.* [8] introduced a LoRa-based smart soldier jacket that employed threshold-based monitoring for panic detection. A predefined pulse-rate threshold of approximately 150 BPM was used to trigger alerts. The study validated long-range LoRa communication and environmental sensing under different conditions; however, decision-making relied solely on fixed thresholds rather than intelligent prediction.

Veerasamy *et al.* [9] presented a LoRaWAN-based soldier health detection and position tracking system using IoT technologies. The system continuously monitored soldier health parameters and generated alerts through automated analysis and emergency-button activation. While effective for monitoring and communication, the approach did not include AI-based predictive assessment.

Singh *et al.* [10] developed an IoT-based soldier tracking and health indication system using Arduino Uno to monitor pulse rate, SpO₂, temperature, and location while providing emergency alerts. Although the system enabled low-cost monitoring and emergency response, it did not employ machine learning for predictive health-risk assessment.

From the reviewed literature, it is evident that existing soldier monitoring systems primarily focus on physiological monitoring, location tracking, communication, and emergency alert generation. Although recent studies [1], [2] highlight AI and machine learning as future directions, none of the reviewed works implement machine learning for predictive soldier health-risk assessment. This research gap motivates the proposed Decision Tree-based system for early risk prediction using physiological, environmental, and motion-sensor data transmitted via LoRa communication.

III. SYSTEM DESIGN AND IMPLEMENTATION

The proposed soldier health monitoring and early risk prediction system is designed using a modular architecture consisting of sensing, processing, communication, and prediction subsystems. The primary objective is to enable real-time risk classification using a machine learning model deployed at the receiver side, thereby reducing computational load on the transmitter and extending operational reliability.

A. Overall System Architecture

The system comprises two main units: a transmitter unit deployed near the soldier and a receiver unit. The transmitter collects physiological, environmental, and motion data using on-board sensors, then sends this raw data via LoRa Ra-02 to the receiver. The receiver runs a pre-trained Decision Tree

model on an ESP32 connected to a laptop, which displays the predicted risk state on a simple web dashboard. No continuous transmission occurs unless data is sampled periodically.

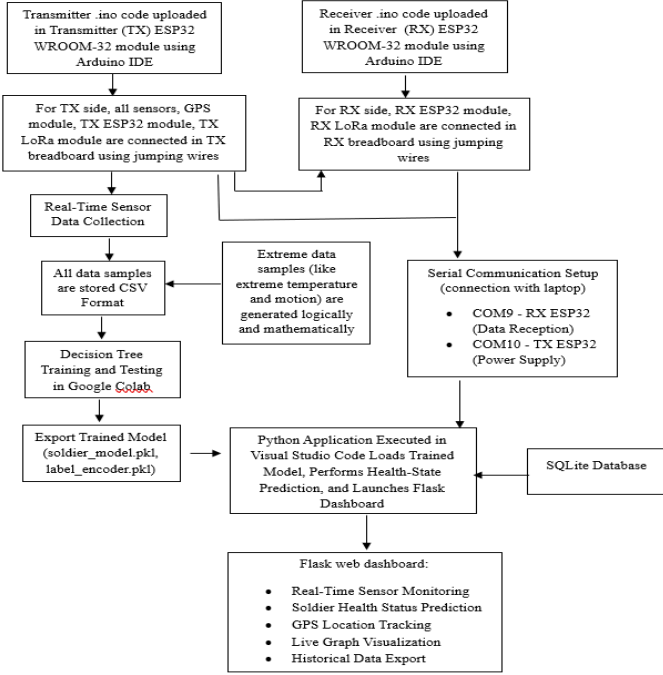


Fig. 1. BLOCK DIAGRAM

In above Fig. 1 represent the block diagram of the projected model.

B. Sensing Subsystem

The transmitter integrates multiple sensors: a MAX30102 sensor for heart rate (BPM) and blood oxygen saturation (SpO_2), a DHT22 sensor for ambient temperature and humidity, and an MPU6050 six-axis accelerometer/gyroscope for motion detection along X, Y, and Z axes. A GPS-NEO 6M module provides location coordinates. These sensors collectively capture all parameters required for the Decision Tree model to classify health states.

C. Processing and Decision-Making Unit

The receiver-side Python application loads the pre-trained Decision Tree model (soldier_model.pkl) and label encoder (label_encoder.pkl) generated during the training phase. The model uses seven input features: BPM, SpO_2 , temperature, humidity, AX, AY, and AZ. Based on these inputs, it classifies the soldier's condition into one of eight health states: Normal, Fatigue, Heat_Risk, Cold_Risk, Panic_Risk, Injury_Risk, Heat_Stroke, or Severe_Cold_Risk. The trained model was developed and evaluated using sensor datasets collected under normal and simulated extreme environmental conditions.

D. Communication Subsystem

The communication subsystem utilizes two LoRa Ra-02 modules operating as transmitter and receiver nodes. Sensor readings and GPS coordinates collected by the transmitter

are transmitted wirelessly through LoRa communication. At the receiver side, the received data are forwarded through serial communication to a laptop running the Python-based monitoring application. The application performs health-state prediction using the trained Decision Tree model and updates the Flask web dashboard in real time.

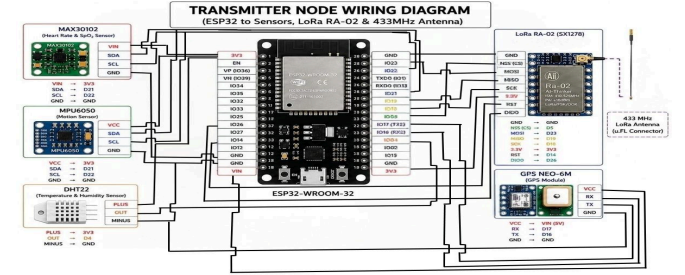


Fig. 2. Circuit diagram of the proposed model, showing the Transmitter Node

In above Fig. 2 represent the Circuit diagram of the Transmitter node (ESP32 with MAX30102, MPU6050, DHT22, GPS, and LoRa Ra-02)

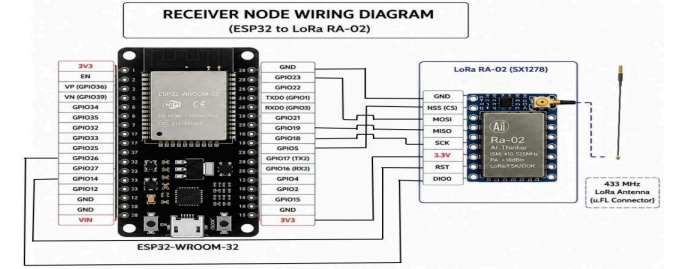


Fig. 3. Circuit diagram of the proposed model, showing the Receiver Node

In above Fig. 3 represent the Circuit diagram of the Receiver node (ESP32 and LoRa Ra-02).

IV. SYSTEM DEPLOYMENT AS A STANDALONE MODEL

A. Model Deployment Design

The proposed system is implemented as a compact standalone hardware prototype and is not directly attached to the soldier's body. The transmitter unit, comprising an ESP32 WROOM-32, MAX30102, MPU6050, DHT22, NEO-6M GPS, LoRa Ra-02 433MHz, and a power source, is arranged on a breadboard and the receiver unit, consisting of a second ESP32 and LoRa Ra-02 module is arranged on a breadboard, is placed at a remote monitoring station and connected to a laptop. The laptop runs the Python-based monitoring application, machine learning inference engine, SQLite database, and Flask web dashboard, enabling real-time visualization of sensor data, health status prediction, and GPS-based location tracking.

B. Component Placement in the Model

Table I presents the hardware components used in the proposed system, their placement in the prototype, and their

respective purposes.

TABLE I. COMPONENTS AND ITS REASON FOR USE

S · N O	COMPONENT	PLACEMENT IN MODEL	REASON
1	Heart Rate / SpO ₂ (MAX30102)	External probe connected to model	Measures BPM and oxygen level
2	Temperature and Humidity (DHT22)	Inside model casing	Captures ambient environmental conditions
3	Motion Sensor (MPU6050)	Inside model casing	Detects vibration/movement along AX, AY, AZ
4	GPS NEO-6M module (with ceramic patch antenna)	Top-facing position	To capture latitude and longitude signals from satellite
5	LoRa Antenna (Ra-02) (2 Units)	External elevated position	Long-range communication reliability between transmitter (TX) and receiver (RX) units
6	ESP32 WROOM-32 Module (2 Units)	Integrated in TX and RX sections	One ESP32 controls the TX side and another controls RX side for data processing and communication management
7	LoRa Module (Ra-02) Module (2 Units)	Connected to TX and RX sections	One LoRa transmits data and other receives data over long distances

C. Motion & Inactivity Interpretation

The MPU6050 sensor provides three-axis acceleration data (AX, AY, AZ) that reflect movement and vibration conditions around the deployed transmitter unit. These measurements are used as indicators of activity levels and environmental disturbances, which contribute to the overall health-state classification process. Motion data are not used independently to determine emergencies; instead, they are analyzed together with physiological parameters such as BPM and SpO₂ to improve the accuracy of risk prediction and reduce false alarms.

V. DECISION TREE-BASED EARLY RISK PREDICTION SYSTEM

To enable early health-risk prediction in soldier monitoring, the proposed system employs a supervised Decision Tree model. The model is trained on a labeled dataset comprising real sensor samples collected from the transmitter under varying hot and cold conditions, together with logically and mathematically generated samples representing extreme conditions that could not be practically reproduced.

A. Motivation for Decision Tree Selection

No previously published IEEE paper in this domain has used any AI or ML method for soldier health risk prediction. Existing systems rely on fixed threshold-based monitoring, which may lead to false alarms or missed detections. A Decision Tree was selected because it provides high interpretability, can effectively handle large labeled datasets with multiple health-related features, requires relatively low computational complexity, and offers fast inference for real-time health-risk prediction before clinical symptoms become severe.

B. Input Features

The Decision Tree uses seven input features:

1. Heart rate (BPM) from MAX30102
2. SpO₂ from MAX30102
3. Ambient temperature from DHT22
4. Ambient humidity from DHT22
5. Motion along X-axis (AX) from MPU6050
6. Motion along Y-axis (AY) from MPU6050
7. Motion along Z-axis (AZ) from MPU6050

These features are transmitted from the transmitter to the receiver via LoRa.

C. Decision Logic and Classification

Table II presents the class labels used for Decision Tree-based health-risk prediction along with the corresponding number of samples and class descriptions.

TABLE II. CLASS LABELS FOR RISK PREDICTION

CLASS LABEL	SAMPLES	DESCRIPTION
Normal	2461	All parameters within safe range
Fatigue	1769	Elevated physiological stress indicator
Heat_Risk	1179	Signs of heat-related stress
Cold_Risk	423	Signs of cold-related stress
Panic_Risk	275	Elevated heart rate with abnormal motion patterns
Injury_Risk	256	Abnormal motion and physiological readings
Severe_Cold_Risk	250	Severe cold-related condition
Heat_Stroke	231	Severe heat-related condition

The Decision Tree model was trained and tested using a labeled dataset comprising real sensor data collected from the transmitter unit and logically generated samples representing extreme environmental and physiological conditions that could not be safely reproduced during experimentation. The logical samples were generated using realistic physiological parameter ranges and environmental reference values to represent extreme operating conditions. These samples were

systematically generated using mathematically consistent combinations of physiological and environmental parameters to represent realistic extreme scenarios while maintaining logical consistency among the input features. The real and generated datasets were then combined to ensure adequate representation of all eight health classes, reduce class imbalance, and improve the model's ability to generalize across normal and extreme operating conditions.

Fig. 4. MLSSHMERPS_Data_Samples.csv

In above Fig. 4 represent the Data samples stored in CSV format to perform training and testing Decision tree in Google Colab.

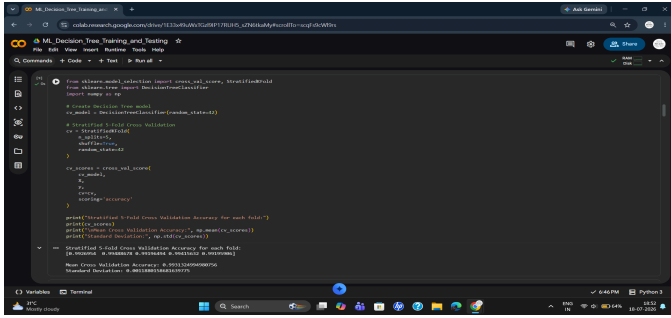


Fig. 5. Cross-validation of Decision Tree in Google Colab

In above Fig. 5 represent the results of the Stratified Five-Fold Cross Validation of the Decision Tree model, which is more appropriate for the multi-class nature of the proposed dataset as it preserves the class distribution across all validation folds, achieving a mean accuracy of 99.31% with a standard deviation of 0.12%, demonstrating consistent model performance across all validation folds.

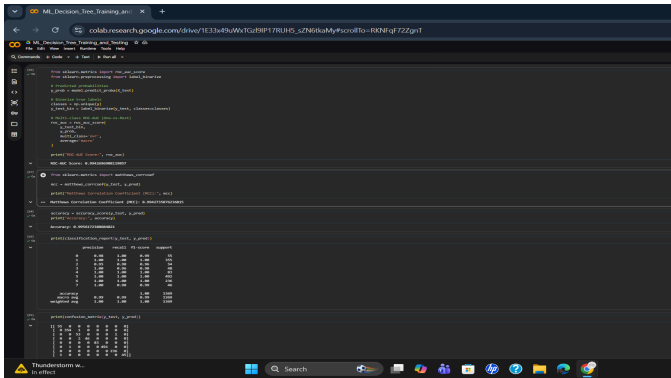


Fig. 6. Decision Tree model evaluation in Google Colab

In above Fig. 6 represent the evaluation results of the Decision Tree model, including the overall accuracy (99.56%),

ROC-AUC score (0.99426), MCC (0.99427), precision, recall, F1-score, and confusion matrix obtained during testing.

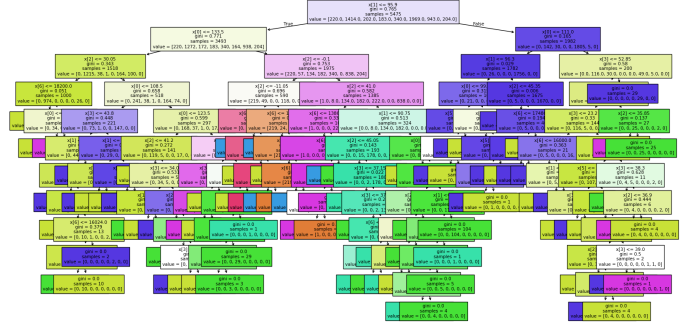


Fig. 7. Complete Decision Tree model

In above Fig. 7 represent the complete Decision Tree model generated during the training and testing phase in Google Colab.

D. Real-Time Monitoring and Status Visualization

The web dashboard continuously displays the current Soldier's Health Status, sensor readings, and GPS coordinates in real time. The predicted health status is displayed based on the Decision Tree classification shown in Table II. No transmission occurs from the receiver to the transmitter, minimizing communication overhead.

VI. HARDWARE IMPLEMENTATION

The hardware implementation is divided into transmitter and receiver sections, selected for availability, low cost, and compatibility with the Decision Tree model.

A. Transmitter Embedded Processing Unit

An ESP32 Dev module serves as the transmitter controller, interfacing with MAX30102, MPU6050, DHT22, GPS-NEO 6M, and LoRa Ra-02. The ESP32 collects sensor data and sends it via LoRa to the receiver. Wi-Fi and Bluetooth are disabled to reduce power.

B. Physiological Sensing Module

MAX30102 measures BPM and SpO₂. These values are critical inputs for detecting fatigue, panic, heat stroke, and severe cold stress.

C. Environmental Monitoring Module

DHT22 measures ambient temperature and humidity — essential for distinguishing heat-related vs. cold-related risks.

D. Motion Detection Module

MPU6050 provides AX, AY, AZ motion data, which are used as input features for the Decision Tree-based health-state classification model.

E. Location Tracking Module

GPS-NEO 6M provides latitude and longitude, transmitted along with sensor data [7].

F. Communication Module (Transmitter and Receiver)

Two LoRa RA-02 modules enable long-range, GSM-free communication. The transmitter sends sensor and GPS data packets to the receiver through LoRa communication.

VII. SOFTWARE IMPLEMENTATION

A. Firmware Architecture

The transmitter firmware handles sensor sampling, packet formation, LoRa transmission, and ACK-based latency measurement, while the receiver firmware handles LoRa reception, ACK generation, and communication metric evaluation.

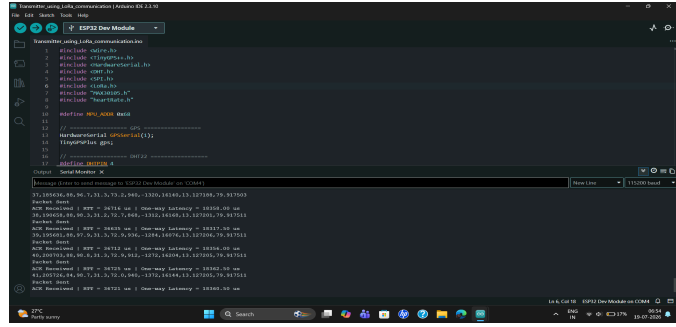


Fig. 8. Arduino IDE showing the transmitter-side firmware code

In above Fig. 8 represent the Arduino IDE showing the transmitter-side firmware code with LoRa communication and ACK-based latency measurement, successfully uploaded to the ESP32 transmitter module.

B. Sensor Data Acquisition and Preprocessing

Sensor readings obtained from MAX30102, DHT22, MPU6050, and NEO-6M are collected and formatted into a unified data packet before transmission.

C. Data Transmission via LoRa

Data packets containing BPM, SpO₂, temperature, humidity, AX, AY, AZ, and GPS are sent periodically from transmitter to receiver.

D. Receiver Decision Tree Execution

The sensor data received through LoRa are processed by a Python-based application, which uses the pre-trained Decision Tree model to predict the soldier's health state.

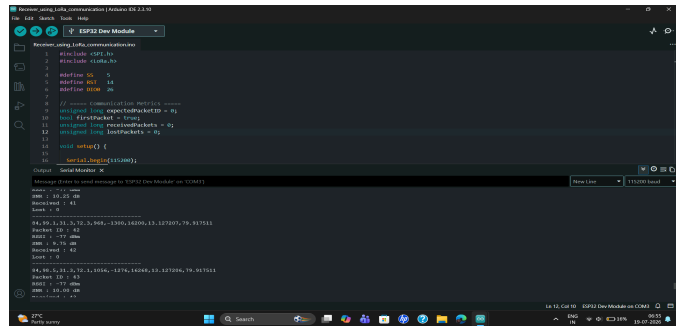


Fig. 9. Arduino IDE showing the receiver-side firmware code

In above Fig. 9 represent the Arduino IDE showing the receiver-side firmware code with RSSI, SNR, packet loss, and ACK generation, successfully uploaded to the ESP32 receiver module.

E. Flask-Based Web Dashboard Display

A Flask-based web dashboard displays real-time sensor readings, machine learning-based health status predictions, and GPS location data. An interactive Leaflet map is integrated to provide live geographical tracking of the soldier based on latitude and longitude provided by the GPS module. If any sensor parameter becomes zero due to a technical fault or communication issue, the system identifies the data as inconsistent and displays the soldier's health status as Data_Inconsistency.

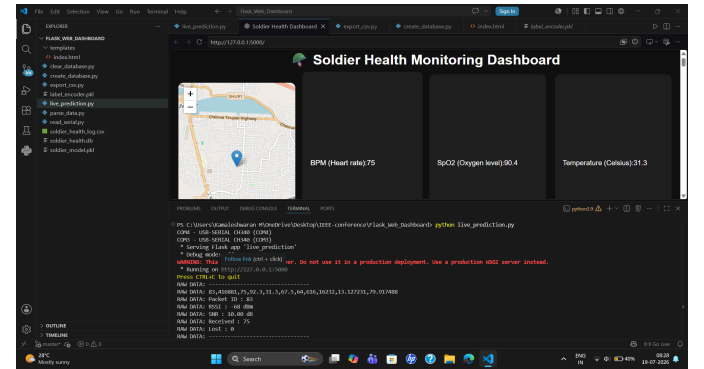


Fig. 10. Visual Studio Code

In above Fig. 10 represent the Real-time Soldier Health Monitoring Dashboard executed through Visual Studio Code and displayed using a Flask-based web interface.

F. Database Management and Historical Health Data Storage

An SQLite database is used to maintain historical records of received monitoring data. The stored information can be exported through the dashboard in CSV format for offline analysis and documentation purposes.

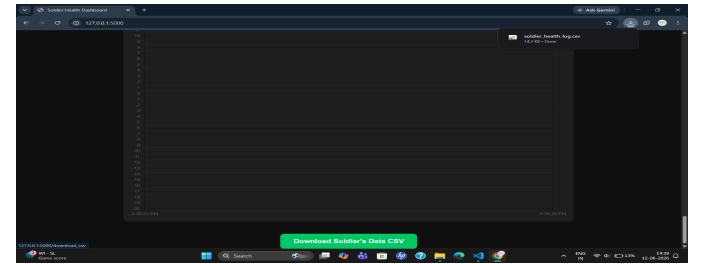


Fig. 11. Downloading historical soldier data in CSV format

In above Fig. 11 represent the downloading historical soldier data in CSV format through the Flask web dashboard.

G. Communication Control

The receiver continuously waits for incoming LoRa packets from the transmitter and forwards the received data to the monitoring application for processing and dashboard updates.

VIII. RESULTS AND DISCUSSION

A. Experimental Observations

A prototype comprising transmitter and receiver units was developed for evaluation. The system integrated physiological, environmental, motion, and location sensors with a microcontroller-based wireless communication setup. The receiver, implemented on a laptop, decoded incoming data and displayed real-time health-state predictions. To evaluate system behavior under varying conditions, the transmitter was tested in controlled outdoor-like environments simulating hot desert (Jaisalmer-like) and cold mountainous (Leh-like) scenarios. Sensor data were continuously transmitted, logged at the receiver, and used for real-time monitoring and model evaluation.

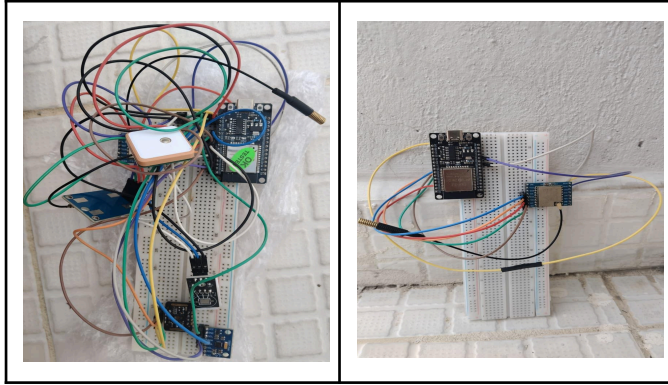


Fig. 12. (a) Transmitter Node and (b) Receiver Node

In above Fig. 12 represent the experimental hardware setups of the (a) transmitter node (left) and (b) receiver node (right), used for data acquisition and wireless communication.



Fig. 13. Hardware prototype setup

In above Fig. 13 represent the transmitter node (COM10, left) consists of an ESP32 interfaced with the MAX30102, MPU6050, DHT22, GPS NEO-6M, and LoRa RA-02 modules. The receiver node (COM9, right) consists of an ESP32 with a LoRa RA-02 module. Sensor data are transmitted wirelessly via LoRa from the transmitter to the receiver. The Python-based monitoring application and Flask web dashboard acquire data exclusively from COM9 (receiver). The COM10 connection is used solely for powering the transmitter node and is not utilized for data acquisition or processing.

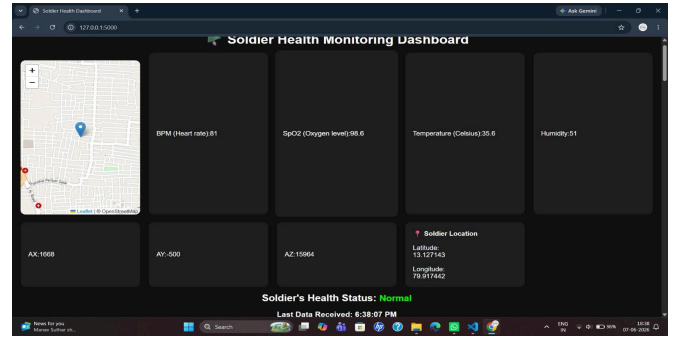


Fig. 14. Flask-based Web dashboard

In above Fig. 14 represent the Web-based dashboard displaying real-time heart rate, SpO₂, temperature, humidity, motion, GPS location, and health status of the soldier using Decision tree prediction.

B. Physiological and Environmental Data Observation

BPM and SpO₂ varied naturally with temperature extremes. The Decision Tree successfully distinguished between heat-related and cold-related stress.

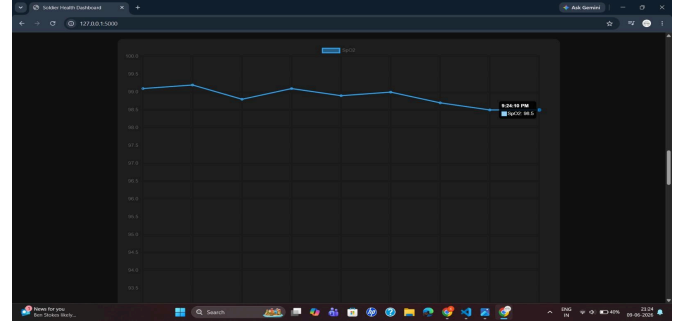


Fig. 15. Real-time graphical trends of SpO₂ vs Time

In above Fig. 15 represent the Real-time graphical trends of SpO₂ vs Time displayed on the Soldier Health Monitoring Dashboard.



Fig. 16. Real-time graphical trends of BPM vs Time

In above Fig. 16 represent the Real-time graphical trends of BPM vs Time displayed on the Soldier Health Monitoring Dashboard.

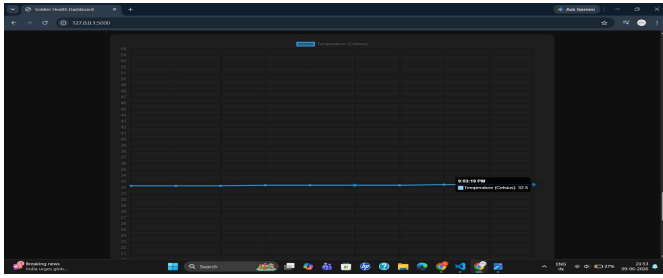


Fig. 17. Real-time graphical trends of environmental temperature (celsius) vs Time

In above Fig. 17 represent the Real-time graphical trends of environmental temperature (celsius) vs Time displayed on the Soldier Health Monitoring Dashboard.

IX. LIMITATIONS

A. No Wearable Implementation

The system is a standalone model, not attached to the soldier, so real-time data may differ from actual moving conditions.

B. Voltage Regulation Not Yet Implemented

No Li-ion battery with voltage regulation is currently used; future work will include buck/boost converters for protection. In addition, a detailed power consumption analysis and battery endurance evaluation have not been performed in the current prototype and will be considered in future work.

C. No Data Encryption Yet

LoRa communication is currently unencrypted, making it susceptible to unauthorized access and data interception. Since the current work focuses on validating the health monitoring and prediction framework, security mechanisms were kept outside the implementation scope. Lightweight encryption algorithms and secure authentication mechanisms will be implemented in future work to improve communication security.

X. CONCLUSION AND FUTURE WORK

This paper presented a real-time soldier health monitoring and early risk prediction system based on a Decision Tree classifier. The system integrates physiological, environmental, and motion-related parameters—BPM, SpO₂, temperature, humidity, and three-axis accelerometer data—to classify eight distinct health conditions, enabling early identification of potential risk states. The proposed architecture employs ESP32-based transmitter and receiver units with LoRa Ra-02 modules to enable long-range, GSM-independent communication. Sensor data collected at the transmitter are transmitted to the receiver, where machine learning-based inference is performed and visualized through a web-based dashboard for real-time monitoring. The complete pipeline, including data acquisition, model training, and testing, has been successfully implemented using a standalone hardware prototype. The system demonstrates the feasibility of combining IoT sensing, wireless communication, and

lightweight machine learning for soldier health monitoring applications, while also highlighting opportunities for further enhancement. Future work will focus on incorporating a proper low-power battery management system using buck/boost converters between the Li-ion battery and the ESP32/LoRa modules to prevent voltage variation damage. In addition, lightweight encryption algorithms will be implemented to secure soldier data transmitted between the LoRa modules, thereby improving communication security for practical deployment.

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